

Worksheet — 1.5 Legal, Moral, Cultural & Ethical Issues

OCR A Level Computer Science H446 — Component 01, Section 1.5 (1.5.1 legislation; 1.5.2 moral/ethical/cultural/social impact)

Name: ____ Date: ____

Time: ~35 min.

Activity 1 — Match each law to what it covers

Draw a line, or write the matching letter in the box.

Law		What it covers
1. Data Protection Act 2018 (+ GDPR)	☐	A. How public bodies (police, security services) may monitor/intercept communications, with rules on who can authorise it
2. Computer Misuse Act 1990	☐	B. Intellectual property — protects original works, software, music and designs from being copied without a licence
3. Copyright, Designs & Patents Act 1988	☐	C. How organisations collect, store and process personal data about living individuals
4. Regulation of Investigatory Powers Act 2000 (RIPA)	☐	D. Unauthorised access to computer systems — hacking, malware and unauthorised modification

Answers: 1 = _ 2 = 3 = _ 4 =

Activity 2 — Which law applies?

For each scenario, name the relevant **Act**. For any Computer Misuse Act scenario, also state **which of the three offences** applies (Offence 1 = unauthorised access; Offence 2 = unauthorised access *with intent* to commit a further offence; Offence 3 = unauthorised *modification*).

1. A company keeps customers' addresses for ten years after they stop being customers, with no reason to do so. Act: _____
2. A student logs in with a teacher's password "just to look around" the staff system, but changes nothing. Act: ____ Offence (if CMA): _____
3. Someone downloads and shares the latest blockbuster film for free from a torrent site. Act: _____
4. A criminal hacks into a bank's servers in order to transfer money into their own account. Act: ____ Offence (if CMA): _____
5. A local council secretly intercepts a resident's emails without obtaining the proper authorisation. Act: _____

6. An attacker releases ransomware that encrypts a hospital's files and demands payment. Act: _____
Offence (if CMA): _____

Activity 3 — Stakeholders and impacts

Choose **one** ethical issue (e.g. automation in the workplace, mass surveillance, the digital divide, AI bias, offshoring). Write it below, then complete a benefit **and** a drawback for each stakeholder.

Issue chosen: _____

Stakeholder	Benefit (+)	Drawback (-)
Individuals / consumers		
Employees / workers		
Businesses / employers		
Government / wider society		

Activity 4 — Extended-response PLANNING scaffold

Essay question: "The increasing use of **automated decision-making** (e.g. for loan applications, job shortlisting and medical diagnosis) does more harm than good." **Discuss.** (12 marks)

Plan **before** you write. Fill the skeleton with short bullet points. For each point use: *point* → *consequence* → *stakeholder* → *counter-point*. Aim for **3–4+ stakeholders**.

FOR the statement (arguments it does more harm than good) - _____ - _____ - _____

AGAINST the statement (arguments it does more good than harm) - _____ - _____ - _____

Stakeholders I will name: _____

Justified conclusion (must take a side — often with a condition): - _____ - _____

Activity 5 — Short-answer definitions

1. **Personal data:** _____

2. **Data subject:** _____

3. **Copyright** (and how does it differ from a **patent**?): _____

4. **Digital divide:** _____

5. The difference between GDPR and the DPA 2018: _____

Challenge box

A hacker breaks into an online retailer's servers and steals a database of customer card details. The retailer had stored that database **unencrypted** and had kept it for years longer than needed.

(a) Which **two** Acts are relevant here, and **why each**?

(b) Which CMA offence(s) has the hacker committed? Explain.

Answer key

Activity 1

1 = C · 2 = D · 3 = B · 4 = A

Activity 2 — correct Act for each scenario

1. **Data Protection Act 2018 / GDPR** — breaches *storage limitation* (keeping data longer than needed).
2. **Computer Misuse Act 1990 — Offence 1** (unauthorised access; nothing was changed and there was no further intent).
3. **Copyright, Designs & Patents Act 1988** — distributing a copyrighted work without a licence (piracy).
4. **Computer Misuse Act 1990 — Offence 2** (unauthorised access *with intent* to commit a further offence, i.e. theft). *Accept a note that the DPA may also apply to the bank's data handling.*
5. **Regulation of Investigatory Powers Act 2000 (RIPA)** — interception by a public body without proper authorisation.
6. **Computer Misuse Act 1990 — Offence 3** (unauthorised modification — encrypting/altering files).

Activity 5 — model points

1. **Personal data:** information relating to an identifiable living individual (e.g. name, address, ID number, location, online identifiers).
2. **Data subject:** the living individual whom the personal data is about.
3. **Copyright** protects the *expression* of original works automatically (no registration), giving the rights-holder control over copying/distributing/adapting. A **patent** protects an *invention*, is **not** automatic — it must be applied for and granted.
4. **Digital divide:** the gap between those who have access to technology, the internet and the skills to use it, and those who do not (often poorer, rural, elderly or disabled people).

5. **GDPR** is the EU regulation; the **DPA 2018** is the UK law that implements GDPR in domestic law.

Activity 4 — model essay plan

FOR (does more harm than good) - *No empathy/human judgement* → unusual but valid cases get unfairly rejected → harms **individuals/applicants** → but a human can review borderline cases (appeal route). - *Algorithmic bias* → systems trained on biased data reproduce/amplify discrimination → harms **disadvantaged groups & society** → but bias can be audited and corrected, unlike hidden human prejudice. - *"Black box" / unclear accountability* → hard to explain or appeal a decision; unclear who is liable → harms **individuals & regulators** → but explainable-AI and logging can improve transparency.

AGAINST (does more good than harm) - *Speed and scale* → decisions made instantly on huge volumes of applications → benefits **businesses & consumers** (faster service) → but speed is worthless if the decisions are unfair. - *Consistency / no fatigue* → same rules applied to everyone, no tiredness or mood → benefits **society** (fairer in theory) → but only as fair as the data and rules behind it. - *Better outcomes in some domains* → e.g. medical diagnosis can spot patterns humans miss → benefits **patients** → but should support, not replace, the clinician.

Stakeholders: individuals/applicants · businesses/employers · disadvantaged groups · government/regulators · patients.

Justified conclusion (takes a side, with a condition): Automated decision-making does more good than harm **only if** it is used to *support* rather than fully replace human judgement, the training data is checked for bias, decisions are explainable, and there is always a route to human appeal. Without these safeguards, the harms (bias, lack of empathy, unaccountability) outweigh the benefits. A top-band answer reaches a clear, reasoned position rather than just stopping.

Challenge box

(a) Computer Misuse Act 1990 (the hacker gained unauthorised access to the servers) **and** the **Data Protection Act 2018 / GDPR** (the retailer failed in its duty of *integrity & confidentiality/security* by storing data unencrypted, and breached *storage limitation* by keeping it too long). *Two laws can apply to one event.* **(b)** The hacker has committed **Offence 1** (unauthorised access) and **Offence 2** (access *with intent* to commit a further offence — stealing/using the card details). *Accept that copying the data could also engage the DPA against whoever misuses it.*